Algebra Lineal Computacional

Primer Parcial – 16 de mayo de 2024

Ejercicio 1. (2.5 pts) Sean $S = \{(x_1, x_2, x_3, x_4) \in \mathbb{R}^4 : x_1 + x_2 + x_3 + x_4 = 0\}$ y

$$T = \langle (1, 1, a, -3), (1, -1, 0, 0), (0, 0, 1, 1) \rangle$$

subespacios de $\mathbb{R}^4$ y $f : \mathbb{R}^4 \to \mathbb{R}^4$ la proyección ortogonal sobre $S \cap T$.

a) Hallar los valores de a para que $S \cap T = \langle (1, 1, 1, -3), (1, -1, 0, 0) \rangle$.

b) Para los valores de a encontrados en a), calcular $[f]_{\mathcal{E}}$, la matriz de f en la base canónica $\mathcal{E}$ de $\mathbb{R}^4$.

(a) $a = 1$. A ojo $\in S$ $\notin S$

$$T = \langle \underbrace{(1, 1, 1, -3)}_{\in S}, (1, -1, 0, 0), (0, 0, 1, 1) \rangle$$

$$S \cap T = \langle (1, 1, 1, -3), (1, -1, 0, 0) \rangle$$

(b) $a = 1$

$S \cap T$ es ORTOGONAL $\Leftrightarrow$

$$\langle (1, 1, 1, -3), (1, -1, 0, 0) \rangle = 0 \quad \checkmark$$

NORMALIZO

$$\frac{(1, 1, 1, -3)}{\|(1, 1, 1, -3)\|_2} = \frac{(1, 1, 1, -3)}{\sqrt{1^2 + 1^2 + 1^2 + (-3)^2}} = \frac{(1\ 1\ 1\ -3)}{\sqrt{12}}$$

$$\rightarrow \left(1/\sqrt{12}, 1/\sqrt{12}, 1/\sqrt{12}, -3/\sqrt{12} \right)$$

$$\frac{(1, -1, 0, 0)}{\|(1, -1, 0, 0)\|_2} = \frac{(1, -1, 0, 0)}{\sqrt{2}} = \left(1/\sqrt{2}, -1/\sqrt{2}, 0, 0 \right)$$

$$\text{BON}_{S \cap T} = \left\{ \left(1/\sqrt{12}, 1/\sqrt{12}, 1/\sqrt{12}, -3/\sqrt{12} \right), \left(1/\sqrt{2}, -1/\sqrt{2}, 0, 0 \right) \right\}$$

$$P_{SNT} = \sigma_1 \nu_1^t + \sigma_2 \nu^t$$

$$P_{SNT} = \begin{pmatrix} 1/\sqrt{12} \\ 1/\sqrt{12} \\ 1/\sqrt{12} \\ -3/\sqrt{12} \end{pmatrix} \left(\frac{1}{\sqrt{12}}, \frac{1}{\sqrt{12}} \frac{1}{\sqrt{12}}, -\frac{3}{\sqrt{12}} \right) +$$

$$\begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \\ 0 \end{pmatrix} \left(1/\sqrt{2}, -1/\sqrt{2}, 0, 0 \right)$$

$$P_{SNT} = \begin{pmatrix} 1/12 & 1/12 & 1/12 & -3/12 \\ 1/12 & 1/12 & 1/12 & -3/12 \\ 1/12 & 1/12 & 1/12 & -3/12 \\ -3/12 & -3/12 & -3/12 & 9/12 \end{pmatrix}$$

$$+ \begin{pmatrix} 1/2 & -1/2 & 0 & 0 \\ -1/2 & 1/2 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 7/12 & -5/12 & 1/12 & -3/12 \\ -5/12 & 7/12 & 1/12 & -3/12 \\ 1/12 & 1/12 & 1/12 & -3/12 \\ -3/12 & -3/12 & -3/12 & 9/12 \end{pmatrix}$$

Ejercicio 2. (2.5 pts) Sean $\alpha > 0$ y $A(\alpha) \in \mathbb{R}^{4\times 4}$ un conjunto de matrices que dependen de α, dadas por:

$$A(\alpha) = \begin{pmatrix} 1 & 2 & 0 & 1 \\ \alpha & \alpha^2+2\alpha & \alpha & 2\alpha \\ 1 & 2 & \alpha^2-1 & 1 \\ 1 & 2 & 0 & 2 \end{pmatrix}.$$

a) Encontrar los valores de α para los cuales $A(\alpha)$ no es inversible.

b) Probar que $Cond_\infty(A(\alpha)) \xrightarrow[\alpha\to 1]{} +\infty$.

c) ¿Qué puede decir de $\lim_{\alpha\to 1} Cond_2(A(\alpha))$?

$\alpha > 0$, $A(\alpha) \in \mathbb{R}^{4\times 4}$

$$A(\alpha) = \begin{pmatrix} 1 & 2 & 0 & 1 \\ \alpha & \alpha^2+2\alpha & \alpha & 2\alpha \\ 1 & 2 & \alpha^2-1 & 1 \\ 1 & 2 & 0 & 2 \end{pmatrix}$$

(a)

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ \alpha & \alpha^2+2\alpha & \alpha & 2\alpha \\ 1 & 2 & \alpha^2-1 & 1 \\ 1 & 2 & 0 & 2 \end{pmatrix}$$

$E_{12}(-\alpha)$ ⇝ $E_{13}(-1)$ $E_{14}(-1)$

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & \alpha^2 & \alpha & 2\alpha-\alpha \\ 0 & 0 & \alpha^2-1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

$\det(A(\alpha)) = 0 \leftrightarrow A(\alpha)$ NO INVERSIBLE

$\det(A(\alpha)) = 1 \cdot \alpha^2 (\alpha^2-1) \cdot 1 = 0$

$\leftrightarrow \alpha = 1 \quad (\alpha > 0)$

(b) $\lim_{\alpha \to 1} \text{Cond}_\infty(A(\alpha)) = +\infty$

Se que

$$\text{COND}_\infty(A) \geq \frac{\|A\|_\infty}{\|A-B\|_\infty} \quad B \text{ SINGULAR}$$

Elijo B/

$$B = A(1) = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 1 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 \\ 1 & 2 & 0 & 2 \end{pmatrix}$$

$$A-B = \begin{pmatrix} 0 & 0 & 0 & 0 \\ \alpha-1 & \alpha^2+2\alpha-3 & \alpha-1 & 2\alpha-2 \\ 0 & 0 & \alpha^2-1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\lim_{\alpha \to 1} \text{COND}_\infty(A) \geq \lim_{\alpha \to 1} \frac{\|A\|_\infty}{\|A-B\|_\infty}$$

$\to 6$

$$\lim_{\alpha \to 1} \frac{\|A\|_\infty}{\|A-B\|_\infty} = \frac{|\alpha| + |\alpha^2+2\alpha| + |\alpha| + |2\alpha|}{|\alpha^2-1|}$$

$\to 0$

$$= +\infty$$

∴ $\text{COND}_\infty(A(\alpha)) \xrightarrow{\alpha \to 1} +\infty$

(C) ¿ $\lim_{\alpha \to 1} \text{COND}_2(A(\alpha))$?

Se cumple: SEA $A \in \mathbb{R}^{n \times n}$

$$\frac{1}{\sqrt{n}} \|A\|_\infty \leq \|A\|_2 \leq \sqrt{n}\, \|A\|_\infty$$

$$\frac{1}{\sqrt{n}} \|A^{-1}\|_\infty \leq \|A^{-1}\|_2 \leq \sqrt{n}\, \|A^{-1}\|_\infty$$

$$\text{COND}_2(A(\alpha)) = \|A(\alpha)\|_2 \|A(\alpha)^{-1}\|_2$$

ENTONCES $(A = A(\alpha))$ SIMPLIFICO

$$\frac{1}{\sqrt{4}} \cdot \frac{1}{\sqrt{4}} \|A\|_\infty \|A^{-1}\|_\infty \leq \|A(\alpha)\|_2 \|A(\alpha)^{-1}\|_2$$

$$\frac{1}{4} \|A\|_\infty \|A^{-1}\|_\infty \leq \text{COND}_2(A(\alpha))$$

$$\frac{1}{4} \text{COND}_\infty(A(\alpha)) \leq \text{COND}_2(A(\alpha))$$

$$\lim_{\alpha \to 1} \underbrace{\frac{1}{4}}_{\frac{1}{4}} \cdot \underbrace{\text{COND}_\infty(A(\alpha))}_{\to \infty} \leq \lim_{\alpha \to 1} \text{COND}_2(A(\alpha))$$

$\lim_{\alpha \to 1} C_{DNO_2}(A|\alpha) \geq +\infty$

Ejercicio 3. (2.5 pts)

a) Sea $A \in \mathbb{R}^{3\times 3}$ una simétrica y definida positiva, probar que

i) Todos sus autovalores son reales y positivos.

ii) El $\det(A)$ es igual al producto de los autovalores de A.

(ii) A SIMETRICA Y REAL $\rightsquigarrow$ DIAGONALIZABLE

$$\det(A) = \det(C D C^{-1})$$

$$= \det(C)\det(D)\det(C^{-1})$$

$$= \det(C)\det(D)\frac{1}{\det(C)}$$

SIENDO D LA MATRIZ DIAGONAL DE LOS AUTOVALORES DE A /

$$\det(D) = \prod_{i=1}^{n} \lambda_i = \prod_{i=1}^{n} d_{ii}$$ ■

(i) $\forall\, v \neq 0 .\ v^t A v > 0$

v_i AUTOVECTOR

$$v^t \lambda v > 0$$

$$\rightsquigarrow \underbrace{\lambda}_{>0}\underbrace{v^t v}_{>0} > 0 \rightsquigarrow \underbrace{\lambda}_{>0}\underbrace{\|v\|^2}_{>0} > 0$$

ii) El $\det(A)$ es igual al producto de los autovalores.

b) Considere la matriz $A = \begin{pmatrix} 7 & 1 & -2 \\ 1 & 7 & -2 \\ -2 & -2 & 10 \end{pmatrix}$,

i) Usando que tiene un autovalor doble y que 12 es autovalor simple, calcular todos los autovalores de A.

ii) Probar que A es diagonalizable.

i) $Av = \lambda v \rightarrow (A - \lambda I)v = 0$

$\leftrightarrow \det(A - \lambda I) = 0$

$$\det \begin{vmatrix} 7-\lambda & 1 & -2 \\ 1 & 7-\lambda & -2 \\ -2 & -2 & 10-\lambda \end{vmatrix}$$

$$= (7-\lambda)(-1)^{1+1} \det \begin{vmatrix} 7-\lambda & -2 \\ -2 & 10-\lambda \end{vmatrix}$$

$$+ (1)(-1)^{2+1} \det \begin{vmatrix} 1 & -2 \\ -2 & 10-\lambda \end{vmatrix}$$

$$+ (-2)(-1)^{3+1} \det \begin{vmatrix} 1 & -2 \\ 7-\lambda & -2 \end{vmatrix}$$

$$= (7-\lambda)\left[(7-\lambda)(10-\lambda) - 4\right]$$

$$+ (-1)\left[(10-\lambda) - 4\right]$$

$$+ (-2)\left[-2 - (-2)(7-\lambda)\right]$$

$$= (7-\lambda)[70 - 7\lambda - 10\lambda + \lambda^2 - 4]$$

$$+ (-1)[6 - \lambda] + (-2)[-2 - (-14 + 2\lambda)]$$

$$= (7-\lambda)[-17\lambda + \lambda^2 + 66]$$

$$+ (-6 + \lambda) + (-2)[-2 + 14 - 2\lambda]$$

$$= -119\lambda + 7\lambda^2 + 462$$

$$+ 17\lambda^2 - \lambda^3 - 66\lambda$$

$$- 6 + \lambda + 4 - 28 + 4\lambda$$

$$= -\lambda^3 + 24\lambda^2 - 180\lambda + 432$$

$$-\lambda^3 + 24\lambda^2 - 180\lambda + 432 \quad | \quad \lambda - 12$$

$$-\lambda^3 + 12\lambda^2 \qquad\qquad -\lambda^2 + 12\lambda - 36$$

$$12\lambda^2 - 180\lambda + 432$$

$$12\lambda^2 - 144\lambda$$

$$-36\lambda + 432$$

$$-36\lambda + 432$$

$$0$$

$$= (\lambda - 12)(-\lambda^2 + 12\lambda - 36)$$

$$= (\lambda - 12)(\lambda - 6)^2$$

$$\therefore \lambda \in \{12, 6\}$$

(ii) Do it!